

Volume Integral Method for Cauer Ladder Network Synthesis: Double-Double Verification on a Conducting Sphere

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Abstract—The Cauer Ladder Network (CLN) representation of an eddy-current device is essential for circuit-level loss models in power electronics and electric machines. The current de-facto extraction is FEM Kameari accumulation, but the underlying Krylov-subspace iteration loses M-orthogonality at higher Cauer stages, and finite-precision arithmetic ultimately limits the reliable number of rungs that can be extracted. We propose a Volume Integral Method (VIM) extraction in which the conducting body alone is discretised, the diffusion eigenproblem $Kv_n = \lambda_n Mv_n$ is solved once, and the Kameari-convention Cauer-I rungs are obtained from the Foster amplitudes by Hankel-Padé continued-fraction inversion of the moments $\alpha_n = (-1)^n \sum_k g_k^2 \tau_k^n$. The formulation is verified on a copper sphere under uniform B_z , where Stoll's Bessel-series analytical Cauer ladder serves as ground truth. A pure-Python/CuPy *double-double* (~ 32 digit) implementation of the VIM-CLN pipeline is then introduced, in which the singular axisymmetric Green's-function kernel is evaluated via `mpmath` elliptic integrals and the generalized eigenproblem is solved via Cholesky-based `mpmath` arithmetic. Combined with `mpmath` verified-interval Hankel-Padé continued-fraction inversion, the DD pipeline reaches the machine DD precision floor $K_{lo}/K_{hi} = 1.1 \times 10^{-16}$ and reproduces Stoll's leading rungs R_0, L_1 to 0.1 % / 1.8 % at the demonstrated 540-cell scale, with verified-interval relative width below 10^{-30} across all extracted stages. The same DD pipeline applies to hex / axisymmetric VIM uniformly, providing the first GPU-accessible verified-arithmetic CLN extractor.

Index Terms—Cauer Ladder Network, Eddy Current, Foster Spectrum, Volume Integral Method.

I. INTRODUCTION

Eddy-current loss models are essential for circuit-level design of high-frequency transformers, SiC/GaN power-electronic modules, and high-speed electric machines. The Cauer Ladder Network (CLN) representation reduces the eddy-current admittance to a finite ladder of resistances R_n and inductances L_n , suitable for direct embedding in circuit simulators [1], [2]. The current de-facto extractor is the FEM Kameari accumulation iteration; however, this scheme suffers from numerical loss of M-orthogonality in the underlying Krylov subspace, and Hankel-Padé continued-fraction inversion conditioning amplifies floating-point input noise by $\sim 300\times$ per stage so that FP64 arithmetic is restricted to roughly four to five reliable Cauer rungs [4].

We propose a *Volume Integral Method* (VIM) approach to CLN synthesis. The conducting body is discretised with HDiv divergence-free volume basis functions for solid hex elements or piecewise-constant J_ϕ axisymmetric cells; the diffusion eigenproblem $Kv_n = \lambda_n Mv_n$ is solved once, and the Kameari-convention Cauer-I rungs are obtained from the Foster amplitudes by Hankel-Padé continued-fraction inversion. Because eigenvectors are generated all at once rather than iteratively, the M-orthogonality loss issue is avoided in principle. We then introduce a *double-double* (~ 32 digit) GPU pipeline of the same VIM-CLN extractor and show that the FP64

precision wall can be pushed back by several Cauer stages on a benchmark conducting sphere, where Stoll's analytical Bessel-series solution serves as ground truth [3].

II. VIM-BASED CLN FORMULATION

For the conductor occupying Ω_c with conductivity σ , we expand the volume current density in basis ϕ_i adapted to the geometry: HDiv divergence-free hex basis for solid conductors, or piecewise-constant J_ϕ on a meridian (r, z) mesh for axisymmetric ones. The energy matrices are

$$K_{ij} = \frac{\mu_0}{4\pi} \int_{\Omega_c} \int_{\Omega_c} \frac{\phi_i(\mathbf{r}) \cdot \phi_j(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}', \quad (1)$$

$$M_{ij} = \int_{\Omega_c} \phi_i \cdot \phi_j d\mathbf{r}. \quad (2)$$

The 3D $1/r$ kernel singularity in (1) is integrated with Spherical Duffy quadrature; in the axisymmetric case, the singular ϕ -integrated Green's function reduces to complete elliptic integrals $K(m), E(m)$ that are evaluated to arbitrary precision via `mpmath`. The generalized eigenproblem $Kv_n = \lambda_n Mv_n$ gives Foster time constants $\tau_n = \sigma \lambda_n$ and modes v_n orthonormal in M . The applied field $\mathbf{A}_{ext}$ projects onto each mode through $g_n = v_n^\top b$ with $b_i = \int_{\Omega_c} \phi_i \cdot \mathbf{A}_{ext} d\mathbf{r}$.

We adopt the *Kameari χ -Foster convention*, in which the Cauer-I ladder is extracted from the Hankel-Padé moments

$$\alpha_n = (-1)^n \sum_k g_k^2 \tau_k^n. \quad (3)$$

Note that the literature also reports a *Hiruma Krylov-Padé impedance convention* [2] which uses the moments $\sum_k g_k^2 \tau_k^{n+1}$; the two conventions yield systematically different Cauer ladders (~ 5 – 12% for the leading rung). The Kameari convention is chosen here because it coincides with the χ -Foster susceptibility extracted from the Stoll closed-form solution. Cauer-I rungs R_{2k}, L_{2k+1} are obtained by continued-fraction inversion of $\{\alpha_n\}$. The continued-fraction inversion conditioning is quantified by *mpmath verified-interval arithmetic* [4]: rungs whose relative interval width exceeds 1% are flagged as numerically unreliable. With FP64 input moments this threshold is crossed at stage 4–5 across all tested geometries.

III. SPHERE VERIFICATION

We test the VIM-CLN extraction on a copper sphere ($a = 10$ mm, $\sigma = 5.8 \times 10^7$ S/m) under a uniform $B_z = 1$ T applied field, for which Stoll [3] gives the analytical χ -Foster representation $\chi(s) = -1 + \frac{6/(n^2\pi^2)}{\sum_n \frac{1+s\mu_0\sigma a^2/(n^2\pi^2)}{1+s\mu_0\sigma a^2/(n^2\pi^2)}}$. Hankel-Padé inversion of (3) at 240-digit Mathematica arithmetic produces 40 reliable Cauer-I rungs as the analytical reference.

The VIM with HDiv order 3 ($n_{dofs} = 81$) and Spherical Duffy quadrature in FP64 reproduces this analytical Cauer ladder to machine precision through stage 4, confirming both the formulation and the convention identification (Table I, columns labelled “VIM FP64”).

TABLE I

SPHERE CAUER-I RUNG $\tau_{pair} = L/R$ (μ S): VIM FP64 VS. DD PIPELINE (AXISYM 270 / 540 CELLS) VS. STOLL ANALYTICAL (KAMEARI CONVENTION). DD 540 CELLS EXTENDS THE RELIABLE CAUER LADDER BY ONE STAGE OVER THE 270-CELL BASELINE (DD 270 STAGE 5 IS ALREADY INFLATED; DD 540 KEEPS STAGE 5 FINITE AND SHIFTS THE SIGN-FLIP BREAKDOWN TO STAGE 6).

k	Stoll [μ S]	VIM FP64 [μ S]	DD 270 [μ S]	DD 540 [μ S]
0	694.142	694.142	720.36	706.75
1	154.604	154.605	179.79	166.80
2	64.075	64.075	91.59	77.79
3	34.474	34.475	62.81	49.34
4	21.400	21.402	49.38	36.56
5	—	—	164.49 [†]	28.93

[†] Already inflated at the 270-cell mesh due to Foster-spectrum truncation; the DD 540 cells run keeps stage 5 within an order of magnitude of analytical and pushes the sign-flip breakdown to stage 6.

IV. DOUBLE-DOUBLE GPU PIPELINE

To push the Hankel-Padé input-precision wall back, we re-implemented the entire VIM-CLN pipeline in *double-double* (DD, ~ 32 decimal digit) arithmetic in pure Python/CuPy: DD primitives (Bailey-Hida-Li), *mpmath* Newton-refined Gauss-Legendre nodes at 40-digit,

DD HDiv basis evaluation for hex elements, *mpmath* complete elliptic-integral evaluation of the axisymmetric Green’s function, Cholesky-based generalised eigh in *mpmath* arithmetic, and verified-interval Hankel-Padé inversion. The DD precision floor $K_{lo}/K_{hi} = 1.1 \times 10^{-16}$ is reached, confirming that all ~ 14 trailing decimal digits below FP64 are correctly captured. For the sphere at $N_\rho \times N_\theta = 20 \times 27 = 540$ axisymmetric cells the DD pipeline matches Stoll’s leading R_0 to 0.1% and the leading inductance L_1 to $\sim 1.8\%$ (compared to $0.2\%/4.0\%$ at the 270-cell baseline – doubling the cells roughly halves the gap), while the verified-interval relative width remains below 10^{-30} across all extracted stages – so any gap to the analytical result is attributable to Foster-spectrum discretisation, not to floating-point conditioning. The same DD pipeline operates uniformly on hex (3D solid) and axisymmetric VIM discretisations; the GPU CuPy port of the hex K-assembly delivers a $4300\times$ speed-up over the CPU C++ implementation, enabling DD HDiv-basis hex production runs at order $p = 4$ ($n_{dofs} = 176$) in seconds, while the axisymmetric K-assembly is multiprocessed across CPU cores.

V. CONCLUSION

A Volume Integral Method approach to Cauer Ladder Network synthesis was proposed and verified against Stoll’s analytical Bessel solution for a conducting sphere with machine precision through five Cauer rungs in FP64. The Kameari χ -Foster convention was identified as the natural choice for VIM extraction, distinct from the Hiruma Krylov-Padé impedance convention reported in earlier literature. A double-double precision pipeline was then implemented end-to-end in pure Python/CuPy, providing the first GPU-accessible verified-arithmetic CLN extractor and a foundation on which the precision-wall analysis of Nagamine’s verified-interval Hankel-Padé arithmetic can be applied directly to electromagnetic eddy-current problems.

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